

TAYLOR'S UNIVERSITY
SCHOOL OF COMPUTER SCIENCE

COMPREHENSIVE WEEKLY ACTION PLAN

Enhanced with Videos, Animations, Guest Sessions, Quizzes, Case Studies & Research

CSC60904 Deep Learning

Semester 2026

This document provides a detailed week-by-week action plan for three modules derived from the official Module Information (MI). Each week is enhanced with:

Quizzes: Formative & graded MCQ quizzes for continuous assessment.

Case Studies: Nepal-grounded real-world scenarios for in-class discussion.

Research Tasks: Structured literature reading and independent paper search.

Videos & Animations: Curated 3Blue1Brown, StatQuest, Computerphile, Sentdex, Corey Schafer, Two Minute Papers, Google AI, and other high-quality animated explainers.

Guest Sessions: Industry practitioners, researchers, alumni, and domain experts invited for specific weeks.

Reflective Logs: Self-directed learning journals aligned to TGC 4/6/7.

MODULE 1: CSC60904 – DEEP LEARNING (14 Weeks)

Module Leader: Anmol Adhikari | Year 2 | 4 Credits | March | MLO1(C1,30%) MLO2(C3,30%) MLO3(P3,40%)

Assessment: Midterm 30%(Wk9) | Individual 30%(Wk13-14) | Group 40%(Wk6-12)

Refs: [R1]Ugail23 [R2]Vasudevan22 [R3]Geron19 [R4]Goodfellow16 [R5]Charniak19 [R6]Ravichandiran19

Wk	Topics & Content	In-Class Activities	Quiz / Case Study / Research	Videos & Animated Resources	Guest Session	Assessment Due	MLO Refs
1	Intro to NN & DL; Neurons; ANNs	Lecture + Practical: Single neuron in Python	QUIZ 1: Diagnostic pre-test (ungraded). CASE STUDY: Nepal earthquake alert — why AI? RESEARCH: Read R1 Ch.1	3Blue1Brown: But what is a Neural Network? (Ch.1). Animated intro to neurons and layers.	—	Briefing on all assignments	MLO1 R1 Ch.1, R4 Ch.1
2	Shallow vs Deep NNs; CNNs intro; Famous architectures	Lecture + Practical: Explore CNN layers in Keras	QUIZ 2: NN fundamentals MCQ (5 min). CASE STUDY: Devanagari digit recognition — why CNN? RESEARCH: Read R1 Ch.3-4	3Blue1Brown: Gradient descent, how NNs learn (Ch.2). Computerphile: Inside a Neural Network. Two Minute Papers: CNN breakthroughs.	—	Video 1 & 2 submission	MLO1 R1 Ch.3-4, R3 Ch.14

Wk	Topics & Content	In-Class Activities	Quiz / Case Study / Research	Videos & Animated Resources	Guest Session	Assessment Due	MLO Refs
3	Deep Conv Models; ResNets; 1x1 Conv; Inception	Lecture + Practical: Visualise ResNet skip connections	CASE STUDY: ImageNet timeline — how ResNet changed everything. RESEARCH: Read He et al. 2016 (ResNet summary). GROUP: Discussion	3Blue1Brown: Backpropagation (Ch.3). Computerphile: ResNets explained. AI Coffee Break: Inception Network animated.	—	—	ML01 R4 Ch.9, He 2016
4	Keras & DL Libraries; Regression & Classification in Keras	Lecture + Practical: Build first Keras classifier	QUIZ 3: CNN architecture MCQ (graded, 10 min). RESEARCH: Explore keras.io independently. GROUP: Initialise group assignment	Sentdex: Keras with TensorFlow tutorial series (Parts 1-3). DeepLizard: Keras Sequential model animated.	—	MCQ Quiz (ML01). Group init	ML02 R3 Ch.10, R6 Ch.3
5	Backpropagation Part 1; Training NNs; Sigmoid; Activations; Keras Workflow	Lecture + Practical: Step-through backprop by hand + Keras	CASE STUDY: Vanishing gradients in Nepali OCR. RESEARCH: Read R4 Ch.6. GROUP: Briefing + Jupyter notebook	3Blue1Brown: Backpropagation calculus (Ch.4). StatQuest: Backpropagation clearly explained. Welch Labs: Learning to See (animated).	—	Group Assignment briefing	ML02 R4 Ch.6, R6 Ch.4

Wk	Topics & Content	In-Class Activities	Quiz / Case Study / Research	Videos & Animated Resources	Guest Session	Assessment Due	MLO Refs
6	Backpropagation Part 2; Implementing NN in Keras	Lecture + Practical: End-to-end Keras pipeline	QUIZ 4: Backprop & activations (graded, 10 min). CASE STUDY: Bias in Nepal crop training data. GROUP: Video submission	DeepLearning.AI: Andrew Ng — Backprop intuition (C1W3). CodeEmporium: Backpropagation animated step-by-step.	GUEST 1: Industry AI Engineer (TBA) — 'How we use DL in production in South Asia' (30 min talk + Q&A)	Group Video 1&2 submission	ML02 R3 Ch.11, R6 Ch.5
7	NN Optimizers; Regularization (Dropout, L2); Data Shuffling; Transforms	Lecture + Practical: Compare SGD vs Adam on MNIST	RESEARCH: Read Dropout paper (Srivastava 2014, abstract + results). CASE STUDY: Overfitting on small Nepali datasets. REFLECTIVE LOG 1: What optimizer concepts did you self-learn?	StatQuest: Regularization clearly explained. Deeplizard: Dropout animated. Weights & Biases: Hyperparameter tuning animated.	—	Individual assignment progress	ML02 R4 Ch.7-8, R3 Ch.11
8	Independent Learning Week	No class — self-directed study	RESEARCH: Find ONE recent DL paper (2022-2026) for your assignment. Write 200-word summary. MIDTERM	MIT OpenCourseWare: Intro to Deep Learning (full lecture). Yannic Kilcher: How to read a ML paper	—	Ind. assignment discussion; Midterm revision	ML01 ML02 Student-selected

Wk	Topics & Content	In-Class Activities	Quiz / Case Study / Research	Videos & Animated Resources	Guest Session	Assessment Due	MLO Refs
			REVISION: Compile Wk1-7 notes.	(methodology video).			
9	RNNs; GRU; LSTM; Seq2Seq; Attention mechanism	Lecture + Practical: Build simple LSTM text generator	QUIZ 5: Pre-midterm practice (self-marked). CASE STUDY: Nepali language chatbot — why RNN? MIDTERM TEST in class	StatQuest: RNN clearly explained. 3Blue1Brown (collab): Attention in Transformers. Computerphile: LSTM animated.	—	MIDTERM TEST (MLO1, 30%)	MLO2 MLO3 R1 Ch.7, R5 Ch.8
10	Gradient Descent; Local Minima; Vanishing & Exploding Gradients	Lecture + Practical: Visualise loss landscapes in 3D	CASE STUDY: Training instability for Nepal air quality model. RESEARCH: Read R4 Ch.8.	Two Minute Papers: Why deep learning works (animated). DeepLearning.AI: Andrew Ng — Vanishing gradients (C2W1).	GUEST 2: University ML Researcher — 'Current DL research frontiers and Nepal opportunities' (30 min)	Individual assignment progress	MLO2 MLO3 R4 Ch.8
11	Face Recognition; Neural Style Transfer; ConvNets Learning; Cost Functions	Lecture + Practical: Neural style transfer demo	QUIZ 6: RNN/LSTM + Gradients (graded, 10 min). CASE STUDY: Ethics of face recognition in Nepal — surveillance vs	Computerphile: Neural Style Transfer. Two Minute Papers: Face generation evolution. Yannic Kilcher: FaceNet explained.	—	MCQ Quiz + Videos (MLO3)	MLO3 R1 Ch.8, R4 Ch.14

Wk	Topics & Content	In-Class Activities	Quiz / Case Study / Research	Videos & Animated Resources	Guest Session	Assessment Due	MLO Refs
			security. RESEARCH: Watch 3 assigned videos				
12	Intro to NLP; Four stages of NLP; Challenges	Lecture + Discussion: NLP in Nepali context	CASE STUDY: Nepali sentiment analysis — low-resource language challenges. RESEARCH: Read one NLP paper on Nepali text.	Google AI Blog: Understanding NLP (animated). Jay Alammar: The Illustrated Transformer. Computerphile: Natural Language Processing.	GUEST 3: NLP Practitioner / Nepali Tech Startup founder — 'Building NLP products for Nepali market' (30 min + Q&A)	Individual assignment progress	MLO3 R5 Ch.10-11
13	Fundamentals of ChatBot; Training house AI; Custom chatbot	Lecture + Practical: Build rule-based chatbot in Python	CASE STUDY: ChatGPT vs rule-based — when is DL overkill? PRESENTATION: Individual assignment in class. REFLECTIVE LOG 2: Semester learning reflection	Sentdex: Chatbot with Python tutorial. Two Minute Papers: ChatGPT explained (animated evolution).	GUEST 4 (optional): Alumni in AI Industry — 'My career path from Taylor's to AI engineer' (20 min)	INDIVIDUAL ASSIGNMENT PRESENTATION (MLO2)	MLO3 R5 Ch.12, R6 Ch.10
14	Submission week — no class	—	Final report writing. REFLECTIVE LOG 3: Final reflection on lifelong	Review: 3Blue1Brown full Neural Networks playlist (revision). Recap all	—	INDIVIDUAL ASSIGNMENT REPORT (MLO1 MLO2, 30%)	All refs

Wk	Topics & Content	In-Class Activities	Quiz / Case Study / Research	Videos & Animated Resources	Guest Session	Assessment Due	MLO Refs
			learning + self-assessment.	Computerphile DL videos.			

SUMMARY OF ENHANCEMENTS ACROSS ALL THREE MODULES

Activity	Deep Learning (CSC60904)
Formative/Graded Quizzes	6 quizzes (Wk 1,2,4,6,8,11)
Nepal Case Studies	12 cases across 14 weeks
Research Tasks	8 reading tasks + 1 independent paper search
Videos & Animations	28 curated videos (3Blue1Brown, StatQuest, Computerphile, DeepLearning.AI, Sentdex, Two Minute Papers)
Guest Sessions	4 guests (Industry AI Engineer, ML Researcher, NLP Startup Founder, Alumni)
Reflective Logs (TGC 4)	3 logs (Wk 7, 13, 14)
New Topics Added	None (all from MI)
Total Video Minutes (est.)	~5.5 hours recommended viewing

All videos are freely available on YouTube or official platforms. The lecturer should create a shared playlist for each module on the LMS (TMeS) and assign specific videos as pre-class viewing or revision material. Guest session schedules are indicative and subject to speaker availability.

— END OF WEEKLY ACTION PLAN —

Prepared by: Module Leaders | School of Computer Science | Taylor's University